

Dyadic-Shadow Lifting from Good-Scale Möbius Correlations to Almost-Endpoint Prefixes

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July 2026

Abstract

The TPC-149 determinant-two theorem controls periodic two-Möbius correlations on every nonexceptional interval $(N, 2N]$ inside an ambient shell. We lift that estimate to cumulative prefixes by an exact dyadic telescoping. An endpoint is admissible when its first J dyadic ancestors avoid the source exceptional set. The correlation error has no factor J , because the block lengths form a geometric series; the exceptional shadow has only a $J = O(\log \log X)$ measure cost. This proves a genuine almost-endpoint prefix theorem on the actual periodic Möbius core. It does not control every predetermined atomic endpoint.

Keywords. Möbius correlation; cumulative prefix; exceptional scale; dyadic shadow; determinant two.

Claim boundary. The result controls cumulative periodic-core prefixes outside an explicit sparse shadow. Logarithmic density does not prove that a predetermined discrete endpoint avoids that shadow. The multiplier remains bounded periodic. The saving is a power of $\log X$, not a fixed power of X ; no $1/400$, prime-pair lower bound or twin-prime theorem is obtained.

1 Source theorem and notation

Let

$$q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az),$$

where

$$a, s, R \geq 1, \quad d, u \in \mathbb{Z}, \quad (a, s) = 1, \quad as \text{ odd}, \quad su - ad = 2.$$

TPC-149 proves that there are absolute $\eta_0, \kappa_0 > 0$ and, for every sufficiently large X , a set $\mathcal{E}_X^* \subset [\sqrt{X}, X]$ such that

$$\frac{1}{\log X} \int_{\mathcal{E}_X^*} \frac{dN}{N} \ll (\log X)^{-\kappa_0}, \tag{1}$$

and, if $qR \leq (\log X)^{\eta_0}$,

$$\frac{q}{N} \left| \sum_{\substack{z \in \mathbb{Z} \\ N < t(z) \leq 2N}} c_z \rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\kappa_0} \tag{2}$$

for all $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^*$ and every period- R function ρ [1].

2 The dyadic shadow

Fix $A > 0$ and set

$$J = \lceil A \log_2 \log X \rceil, \quad \mathcal{S}_{X,J} = \bigcup_{j=1}^J 2^j \mathcal{E}_X^*. \quad (3)$$

Here $2^j E = \{2^j t : t \in E\}$.

Proposition 2.1 (Shadow measure). *One has*

$$\frac{1}{\log X} \int_{\mathcal{S}_{X,J} \cap [2^J \sqrt{X}, X]} \frac{dT}{T} \ll J(\log X)^{-\kappa_0} = (\log X)^{-\kappa_0 + o(1)}. \quad (4)$$

Proof. For every j , the substitution $T = 2^j N$ preserves logarithmic measure:

$$\int_{2^j \mathcal{E}_X^*} \frac{dT}{T} = \int_{\mathcal{E}_X^*} \frac{dN}{N}.$$

Use the union bound, (1), and $J = O(\log \log X)$. Intersecting with the endpoint interval only decreases the measure. $\square$

3 Cumulative prefix lifting

For bounded period- R ρ , define

$$A_\rho(T) = \sum_{\substack{z \in \mathbb{Z} \\ 0 < t(z) \leq T}} c_z \rho(z).$$

Theorem 3.1 (Almost-endpoint cumulative prefix). *Suppose*

$$qR \leq (\log X)^{\eta_0}, \quad 2^J \sqrt{X} \leq T \leq X, \quad T \notin \mathcal{S}_{X,J}.$$

Then

$$\boxed{\frac{q}{T} |A_\rho(T)| \ll \|\rho\|_\infty \left\{ (\log X)^{-\kappa_0} + 2^{-J} + \frac{q}{T} \right\}.} \quad (5)$$

Consequently,

$$\frac{q}{T} |A_\rho(T)| \ll \|\rho\|_\infty (\log X)^{-\min\{\kappa_0, A\} + o(1)}$$

in the asymptotic determinant-two envelope.

Proof. Use the exact identity

$$A_\rho(T) = \sum_{j=1}^J \sum_{\substack{z \\ T/2^j < t(z) \leq T/2^{j-1}}} c_z \rho(z) + A_\rho(T/2^J). \quad (6)$$

Since $T \notin \mathcal{S}_{X,J}$, every $N_j = T/2^j$ avoids $\mathcal{E}_X^*$. The endpoint assumptions put all N_j in $[\sqrt{X}, X]$, so (2) gives

$$\left| \sum_{N_j < t(z) \leq 2N_j} c_z \rho(z) \right| \ll \|\rho\|_\infty \frac{N_j}{q} (\log X)^{-\kappa_0}.$$

After multiplying by q/T , the sum of these errors is at most

$$\|\rho\|_\infty (\log X)^{-\kappa_0} \sum_{j=1}^J 2^{-j} \leq \|\rho\|_\infty (\log X)^{-\kappa_0}.$$

Thus there is no J loss in the correlation term.

The progression $t(z) = ad + qz$ has at most $T/(q2^J) + 1$ positive points below $T/2^J$. Since $|c_z| \leq 1$,

$$\frac{q}{T}|A_\rho(T/2^J)| \leq \|\rho\|_\infty \left(2^{-J} + \frac{q}{T}\right).$$

Combine the bounds. Finally $2^{-J} \leq (\log X)^{-A}$. $\square$

Corollary 3.2 (Intervals between good endpoints). *If $T_0 < T_1$ both satisfy the hypotheses of [theorem 3.1](#), then*

$$\sum_{T_0 < t(z) \leq T_1} c_z \rho(z) = A_\rho(T_1) - A_\rho(T_0)$$

is bounded by the sum of the two corresponding right sides.

4 Why this is not all-prefix control

Proposition 4.1 (Atomic nonimplication). *The measure estimate [\(4\)](#), by itself, does not imply that a prescribed finite or discrete endpoint set $\mathcal{T}_X$ is disjoint from $\mathcal{S}_{X,J}$.*

Proof. This remains true even if the dyadic form of the shadow is preserved. Indeed, for endpoints in $[2^J\sqrt{X}, X]$, enlarge the source set by

$$\left\{2^{-J}T : T \in \mathcal{T}_X \cap [2^J\sqrt{X}, X]\right\}.$$

This is a finite or countable subset of $[\sqrt{X}, X]$, so it has zero continuous logarithmic measure. Removing these extra scales from the good set cannot invalidate the source estimate, while the new $j = J$ shadow copy contains every displayed endpoint. Thus the same measure bound is compatible with intersection; the measure bound alone supplies no avoidance statement. $\square$

This is the TPC-150 selector firewall in a sharper form [\[2\]](#): the relevant bad set is now the explicit dyadic shadow required by cumulative telescoping. A pointwise theorem, proved exceptional avoidance, or a non-atomic selector crosswalk is still needed for deterministic all-prefix control.

5 Audit and level

The standard-library certificate checks [\(6\)](#) exactly on a rational progression, checks the geometric block mass, and checks a union bound on intervals in logarithmic coordinates. It source-locks the TPC-149 theorem using canonical UTF-8/LF hashes; those hashes certify integrity only.

$$\boxed{\text{TPC-159} = \text{L1_ACTUAL_PREFIX_ALMOST_ENDPOINT.}}$$

This is strictly beyond a single source-native dyadic interval, but below deterministic all-prefix control and below every positive fixed- X -power target.

6 Conclusion

The actual periodic Möbius core now reaches cumulative prefixes on a density-one logarithmic family of endpoints. The dyadic error does not accumulate a $\log \log X$ factor; only the exceptional shadow does. The remaining endpoint problem is no longer vague: one must control the intersection of that shadow with the literal endpoint registry.

References

- [1] Liang Wang. A Small-Polylogarithmic Determinant-Two Möbius Corridor: Uniform Actual-Core Cancellation with Bounded Periodic Data, 2026. TPC-149 manuscript.
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